Mathematics – Elementary Algebra

1 Basic Mathematical Language

1.1 Sets: How to Group Numbers

- Basics: Sets, Subsets, Supersets, Elements
- Set Notation

1.2 Real Numbers and Subsets

1.3 Interval Notation

2 Operations and Variables

2.1 Variables and Constants

- Notation: Subscripts on Variables

2.2 Operations on Variables

2.3 Properties of Operations

- Associativity
  - General, Left, Right, and Nonassociativity
- Commutativity and Anticommutativity
  - Exception Example: Matrix Multiplication
- Relationship between Associativity and Commutativity

2.4 Implied 1

2.5 Order of Operations

2.6 Arithmetic on Like Terms

- Addition/Subtraction
- Multiplication/Division

2.7 Distributive Property

- DP: Multiplication over Addition
- DP: Exponentiation over Multiplication?
- Advanced Topic: General Distributive Property - Left Distribution vs. Right Distribution

3 Expressions and Evaluation

3.1 Expressions

3.2 Evaluating Expressions

3.3 Fractional Expressions

3.4 Simplifying Expressions

3.5 Common Algebraic Errors - Expressions

4 Equations and Inequalities

4.1 Linear Equations and Identities

- Inverse Operations
- Balancing Equations
- Simple Equations
- Constructing Equations from Word Problems
- Consecutive Integers

4.2 Linear Inequalities
- Inequalities with Negative Numbers
- Interval Notation for Solutions
- Graphing Solutions on Number Lines

4.3 Absolute Value Equations and Inequalities

5 Functions and Relations

5.1 Relations
5.2 Types of Relations
- One-to-One
- One-to-Many
- Many-to-One
- Many-to-Many

5.3 Proporties of Relations on a Set
- Reflexivity
- Symmetry / Antisymmetry
- Transitivity

5.4 Functions
5.5 Notation and Interpretation for Functions
- Notation of Functions as Objects vs. Functions of a Variable (f vs. f(x) or f :)
- Rule-First Notation vs. Mapping Notation (f(x) vs. f :)

5.6 Identifying Functions from Relations
- Ordered Pairs
- Graphical Method: Vertical Line Test (VLT)
- Algebraic Method

5.7 Properties of Functions: Domain, Codomain, Range
- Domain Restrictions
- Codomain vs. Range
- Codomains as Part of the Definition of a Function
- Determining Domains and Ranges from Graphs
- Feasible / Appropriate Domains and Ranges from Real-Life Problems

5.8 Properties of Functions: One-to-One, Onto, Invertible
- One-to-One / Injectivity
- Onto / Surjectivity
- Bijectivity and Invertibility

5.9 Properties of Functions: Even and Odd
- Note about Odd Functions: Symmetry and Definedness of Odd Functions at $x = 0$

5.10 Properties of Functions: Increasing and Decreasing, Monotonicity
5.11 Operations on Functions

6 Linear Functions and Inequalities

6.1 2D Cartesian Coordinate System
6.2 Slope/Gradient as Rate of Change
6.3 Equations of a Straight Line
- Relating Slope to Equation
- Relating y-intercepts to Equation
- Forms of Straight Line Equations
  - Point-Slope Form
  - Slope-Intercept Form
  - Normal/Standard Form ($ax + by + c = 0$)
- Conversion between Forms

6.4 Graphing Linear Functions
6.5 Linear Inequalities
- Trichotomy
  - Order Relationships between Numbers: $<$, $\leq$, $>$, $\geq$
  - Strict vs. Weak Inequalities
  - Wrinkle: Truth of Trichotomy with Weak Inequalities?
- Interactions between Operations and Inequalities I: Addition, Subtraction, Multiplication and Division by a Positive Number
- Interactions between Operations and Inequalities II: Multiplication and Division by a Negative Number
- Interactions between Operations and Inequalities III: Reciprocals and Signs
- Solving Linear Inequalities: With Real-Life Problems
- Graphing Linear Inequalities
  - Inequalities of One Variable: Number Line
  - Inequalities of Two Variables: Cartesian Graph
    * Boundary Lines and Half-Planes
    * Solid vs. Dashed Boundary Lines

7 Systems of Linear Equations and Inequalities

7.1 Intuition of Systems of Equations
7.2 Solving Systems of Equations - Substitution
7.3 Solving Systems of Equations - Elimination
7.4 Solving Systems of Equations - Graphing
7.5 Consistency and Dependency - Nature of Solutions
- Consistency
- Dependency

7.6 Systems of Linear Inequalities
- Conjunctions
- Disjunctions
- Applications with Real-Life Problems
- Basic Optimization

== **bookmark**

5 Polynomials

5.1 Monomials and Polynomials
5.2 Arithmetic on Polynomials
- Addition/Subtraction
- Multiplication
- Division
- Multiplication: Multivariate Polynomials
- Division: Multivariate Polynomials

5.3 Side Effects of Division: Polynomial Factor and Remainder Theorems

4 Factoring/Factorization

4.1 Greatest Common Factor, Revisited
4.2 Quadratic Polynomials
4.3 Special Forms: Difference of Two Squares
4.4 Zero Theorem
4.5 Completing the Square
4.6 General Solution to Quadratics: Quadratic Formula

5 Functions and Graphing

5.1 2D Cartesian Coordinate System
5.2 Slope/Gradient
5.3 Equations of a Straight Line
• Relating Slope to Equation
• Relating y-intercepts to Equation
• Forms of Straight Line Equations
• Conversion between Forms
5.4 Graphing Linear Equations
5.5 Relations and Functions
• Relations
• Functions
5.6 Function Testing
• Vertical Line Test (VLT)
5.7 Properties of Functions: Domain, Codomain, Range
• Domain Restrictions
• Codomain vs. Range
• Determining Domain, Codomain and Range from Graphs
5.8 Properties of Functions: Even and Odd
• Note about Odd Functions
5.9 Properties of Functions: Increasing and Decreasing
5.10 Sum/Difference of Functions
5.11 Product of Functions

6 Inequalities

6.1 Trichotomy and Basic Inequalities
6.2 Inequalities and Negative Numbers
6.3 Graphing Inequalities on Number Lines
6.4 Compound Inequalities
• Disjunctions
• Conjunctions

6.5 Graphing Inequalities on Cartesian Planes
6.6 Inequalities with Absolute Values
6.7 Quadratic Inequalities

7 Systems of Linear Equations

7.1 Two-Step Problems
7.2 Solving Systems of Equations - Substitution
7.3 Solving Systems of Equations - Elimination
7.4 Solving Systems of Equations - Graphing
7.5 Number of Solutions
7.6 Solving Systems of Inequalities
• Systems of Quadratic Inequalities

8 Exponents and Radicals II

8.1 Powers of Negative Bases
8.2 Powers of Fractions
8.3 Zero as an Exponent
• Trivia: 0^0?
8.4 Negative Exponents
• Product Rule with Negative Exponents
8.5 Fractional Exponents
8.6 Rationalizing the Denominator
• Conjugates

9 Ratios and Proportions II

9.1 Word Problems
9.2 Applied Mathematics
• Chemical Compounds / Stoichiometry
• Fractions, Decimals, and Percentages
• Percentage Markup/Markdown
• Commissions
• Simple Interest
• Compound Interest
9.3 Complex Fractions
• Ratios and Proportions with Complex Fractions

10 Factoring II

10.1 Factoring Quadratics with a != +/-1
10.2 Factoring by Grouping
10.3 Difference of Two Cubes
10.4 Sum of Two Cubes
• Mnemonic: SOAP - Don't Drop Them!

11 Rational Functions

11.1 Simplifying Rational Functions
11.2 Arithmetic with Rational Functions/Expressions
• Addition/Subtraction
• Multiplication
• Division
  – Domain Restrictions

12 Expressions and Equations II
12.1 Proportionality
• Direct Proportion
• Inverse Proportion
12.2 Fractional Equations
12.3 Rational Equations
12.4 Radical Equations
12.5 Multivariate Equations
12.6 Multivariate Rational Equations

13 Systems of Equations II
13.1 Systems with Subscripts
13.2 Applied Systems of Equations
• Uniform Motion
• Number Word Problems
• Age Word Problems
13.3 Systems of Nonlinear Equations
13.4 Systems of Three Equations

14 Graphing II
14.1 Parabolas
14.2 Parallel and Perpendicular Lines
14.3 Distance between Points
14.4 Circles
• Center and Radius of Circles

15 Functions II
15.1 Operations on Functions, Revisited
• Effects of Operations on Domain and Range
15.2 Composite Functions
• Domains of Composite Functions
• Decomposition
15.3 Function Modelling
15.4 Piecewise Functions
15.5 Horizontal Line Test: Identifying One-to-One Functions
15.6 Inverse Functions
• Domain and Range of Inverse Functions
• Finding Original Functions from Inverses
15.7 Elementary Functions

16 Exponential and Logarithmic Functions
16.1 Exponents and Logarithms
16.2 Common Logarithms and Restricted Values
16.3 Evaluating Exponents and Logarithms
16.4 Solving Exponential and Logarithmic Equations
16.5 Laws of Logarithms
16.6 Euler's Number and Natural Logarithms
16.7 Base Change
16.8 Graphing Exponential and Logarithmic Functions